

Problem set 1

The Poggendorff illusion is a well-known effect where the two visible sections of a partly occluded line do not appear to be in alignment.



In the figure above, for example, the two diagonal lines are perfectly aligned, but to most people the one on the right appears to be shifted downward.

Use Python and PsychoPy to write an experiment to measure the size of this illusion under a few different conditions, with a total of 100 trials.

On each trial, draw the left line, the right line, and the occluding rectangle. The participant uses the mouse to adjust the vertical position of the right line so that it appears to be aligned with the left line. When the participant has made this setting, they press the spacebar, and the next trial begins.

On each trial, show the lines at one of these angles, randomly chosen: -60° , -30° , 0° , 30° , and 60° . Here 0° is horizontal, and positive angles are counterclockwise.

On each trial, show the occluding rectangle at one of three widths, randomly chosen. You can choose values for the three widths.

On each trial, record the following information in a list of lists, with a sublist containing the following information for each trial: (a) trial number, (b) line angle, (c) rectangle width, (d) the participant's match setting, and (e) response time.

[continued on next page]

For this problem set you don't need to do any data analysis.

Here's a useful fact for this problem set. Suppose you have a line segment with length L , at angle θ relative to the x-axis. Suppose one end of the line is at the origin $(0, 0)$. Then the coordinates of the other end of the line are $(L \cos\theta, L \sin\theta)$. If you're unsure how to use this fact, feel free to ask.

Email your solution to me (rfm@yorku.ca) in a single .zip or .tar file named with your last name in lowercase, e.g., murray.zip.

Due October 23, 2026